

Here is the complete text formatted clearly so you can easily copy and paste it into any text editor, Microsoft Word, Google Docs, or export it directly to PDF yourself:

SANDBOX ONLINE ASSESSMENT SYSTEM

Detailed Technical Specification & Team Work Distribution Matrix

1. User Roles & High-Level Architecture

- **Super Admin:** Manage Organizations (Create, Update, Deactivate), Provision HR Accounts, Configure Subscription Plans, and Audit Platform Payments.
- **HR (Organization Admin):** Manage Candidates Directory, Maintain Question Banks (MCQ & Coding), Configure & Publish Assessments, Assign Candidates, and Review Proctoring Reports.
- **Candidate:** Register/Login, Access Portal, Complete Assessments (MCQ & Coding Workspace), and View Immediate Score Breakdowns.

3. Microservices & API Endpoint Specifications

Microservice 1: Authentication & User Service

- **Responsibilities:** JWT Token lifecycle, Spring Security filter chain, BCrypt password encryption, Email verification/resets, Tenant/Organization management, Subscription plans, Razorpay payment processing.
- **Database Tables:** users, roles, organizations, subscriptions, payments
- **API Endpoints:** /auth/**, /organization/**, /subscription/**, /payment/**

Microservice 2: Assessment Engine Service

- **Responsibilities:** Candidate management, MCQ & Coding question bank maintenance, Assessment rules (timers, passing criteria, negative marking), Candidate assignment & email triggers, Monaco Code Editor integration, PDF report generation.
- **Database Tables:** candidates, questions, mcq_options, coding_questions, assessments, assessment_questions, assignments, submissions, coding_submissions, results
- **API Endpoints:** /candidate/**, /question/**, /assessment/**, /assignment/**, /exam/**, /result/**, /report/**

Microservice 3: AI Code Evaluation & Proctoring Service

- **Responsibilities:** Code execution via Judge0 API, Code quality/complexity analysis via OpenAI/Gemini LLM, Browser guardrails (tab switch, window blur, fullscreen exit), 30-second camera snapshot AI vision analysis (face/phone detection), 30-second silent video recording on violation events.

- **Database Tables:** violations, proctoring_logs, ai_evaluations
- **API Endpoints:** /judge/run, /ai/evaluate, /proctoring/snapshot, /proctoring/video, /proctoring/violation

4. Full Team Work Distribution Matrix (6 Members)

Member	Module Scope	Database Tables	Backend Implementation Scope	APIs & Integrations	Frontend & UI Scope
Member 1	Authentication, Security & Organizations	usersrolesorganizations	• JWT Token Service (generation, validation, refresh)• • Spring Security filter chain setup• • BCrypt password encryption pipeline• • Email verification & password reset tokens• • Organization tenant CRUD & lifecycle logic	/auth/**/organization/** <i>SMTP Integration</i>	• Login & Register Pages• • Password Recovery Views• • Super Admin Org Management Dashboard• • HR Account Creation Modal
Member 2	Subscriptions, Payments & Candidate Admin	subscriptionpaymentscandidates	• Subscription tier plan management• • Razorpay payment	/subscription/**/payment/**/candidate/** <i>Razorpay SDK Integration</i>	• Subscription Plans View• • Razorpay Gateway Integration

			- order creation & checkout• - Razorpay webhook listener & verification• - Candidate CRUD business logic & search filter		- UI• Payment Monitoring Dashboard• - HR Candidate Directory UI
Member 3	Question Bank & Assessment Builder	questionsmcq_options coding_questionsassessmentsassessment_questions	• MCQ question CRUD & option validation• - Coding question CRUD with driver code & test cases• - Assessment configuration engine (timer, passing marks, negative marking)• - Assessment publish/unpublish state machine	/question/** /assessment/**	• Question Bank Management UI• - MCQ & Coding Question Form Editors• - Assessment Builder Workflow UI• - Assessment Rules Configurator
Member 4	Exam Engine,	assignmentssubmission	• Candidate assignment	/assignment/**/exam/**/	• Candidate Assessment

	Candidate Portal & Results	nsresults	logic & SMTP notification trigger• Exam state synchronization & auto-save engine• Auto-submit timer listener on backend• Result calculation engine (MCQ + Coding)• PDF Report generation engine	result/**/report/**	t Portal Dashboard• Interactive Examination View (Palette, Timer)• Candidate Result View• Analytics Dashboard & PDF Viewer
Member 5	AI Code Evaluation Engine & Monaco Editor	coding_submissionsai_evaluations	• Judge0 API code submission & execution wrapper• Test case verification pipeline• OpenAI / Gemini LLM prompt engineering for code review• Code quality, complexity & optimizatio	/judge/run/ai/evaluateJudge0 API IntegrationLLM API Integration	• Monaco Code Editor Interface• Language selector & theme switcher• Run Code & Submit Code buttons with test output console• AI Feedback UI Widget

			n analyzer		
Member 6	AI Proctoring & Evidence Logging Service	violationspr octoring_logs	• Browser violation event logger endpoint • 30s camera snapshot AI detection pipeline • Image filtering logic (discard normal, log suspicious) • Silent event video recorder uploader • HR Proctoring Report aggregation API	/proctoring/snapshot/proctoring/video/proctoring/violation <i>AI Vision API Integration</i>	• Hidden browser event listener components • Background WebCam snapshot capture hook • Silent MediaRecorder video capture hook • HR Proctoring Report Dashboard

1. Authentication & Authorization Module

Functionalities

User Registration

- Register Candidate

- Register HR
- Validate Email
- Password Encryption

Login

- JWT Authentication
- Role-Based Login
- Session Management

Password Management

- Forgot Password
- Reset Password
- Change Password

Profile

- View Profile
- Update Profile

Security

- JWT Token Validation
 - Role-Based Access Control
 - Logout
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2. Organization Management Module

Functionalities

- Create Organization
 - Update Organization
 - Delete Organization
 - Activate/Deactivate Organization
 - Manage HR Accounts
 - View Organization Details
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3. Subscription & Payment Module

Functionalities

- View Subscription Plans
 - Purchase Plan
 - Razorpay Payment
 - Verify Payment
 - Subscription Expiry Tracking
 - Renew Subscription
 - View Payment History
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4. Candidate Management Module

Functionalities

HR Side

- Add Candidate
- Edit Candidate
- Delete Candidate
- Search Candidate
- Import Candidate List (CSV - Optional)
- View Candidate Profile

Candidate Side

- Register
 - Update Profile
 - View Assigned Assessments
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5. Question Bank Module

Functionalities

MCQ Questions

- Add MCQ
- Edit MCQ
- Delete MCQ
- Search MCQ
- Categorize Questions
- Set Difficulty Level

Coding Questions

- Add Coding Question
 - Edit Coding Question
 - Delete Coding Question
 - Add Sample Test Cases
 - Add Hidden Test Cases
 - Set Time Limit
 - Set Memory Limit
 - Set Marks
 - Set Difficulty
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6. Assessment Management Module

Functionalities

- Create Assessment
- Edit Assessment
- Delete Assessment
- Add MCQ Questions
- Add Coding Questions
- Configure Timer
- Configure Passing Marks
- Configure Negative Marking
- Configure Start & End Date
- Publish Assessment

- Unpublish Assessment
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7. Candidate Assignment Module

Functionalities

- Select Candidates
 - Assign Assessment
 - Bulk Assignment
 - Send Email Notification
 - View Assigned Candidates
 - Reassign Assessment
 - Remove Assignment
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8. Online Examination Module

Functionalities

Before Exam

- Check Browser Compatibility
- Request Camera Permission
- Request Fullscreen Permission
- Read Exam Instructions

During Exam

- Start Exam
- Countdown Timer
- Next Question
- Previous Question
- Mark for Review
- Question Palette
- Auto Save

- Manual Save
- Auto Submit
- Submit Exam

Supports

- MCQ
 - Coding Questions
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9. Coding Editor Module

Functionalities

- Monaco Editor
 - Language Selection
 - Syntax Highlighting
 - Auto Indentation
 - Run Code
 - Submit Code
 - Display Output
 - Display Compilation Errors
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10. AI Code Evaluation Module

Functionalities

Judge0

- Compile Code
- Execute Code
- Run Hidden Test Cases
- Calculate Score

LLM Evaluation

- Analyze Code Quality
 - Suggest Optimizations
 - Estimate Time Complexity
 - Estimate Space Complexity
 - Generate AI Feedback
 - Generate AI Score
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11. AI Proctoring Module

Functionalities

Browser Monitoring

Silently detect

- Tab Switching
- Window Blur
- Fullscreen Exit
- Copy
- Paste
- Right Click

Store all violations in backend.

Camera Monitoring

- Start Camera
- Capture Snapshot Every 30 Seconds
- Send Snapshot to AI
- Detect Face Presence
- Detect Multiple Faces
- Detect Mobile Phone

If violation detected

- Store Snapshot

- Create Violation Entry

Otherwise

- Discard Snapshot
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Event-Based Video Recording

Trigger Recording On

- Tab Switch
- Fullscreen Exit
- Multiple Faces
- Mobile Phone Detection

Features

- Record 30 Seconds
- Upload Video
- Store Video Path
- Link Video to Violation

Candidate is **not notified** of recording.

Violation Management

Store

- Candidate ID
 - Assessment ID
 - Violation Type
 - Timestamp
 - Snapshot Path
 - Video Path
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12. Result Module

Functionalities

- Calculate MCQ Score
 - Calculate Coding Score
 - AI Evaluation Score
 - Final Score
 - Pass/Fail Status
 - Candidate Result View
 - HR Result View
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13. Report Module

Functionalities

HR Reports

- Assessment Report
- Candidate Performance Report
- Overall Statistics
- Download PDF Report (Optional)

AI Reports

- AI Evaluation Report
- Judge0 Execution Report
- Code Quality Report

Proctoring Reports

- Total Violations
 - Violation Timeline
 - Snapshot Evidence
 - Video Evidence
 - Violation Summary
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14. Notification Module

Functionalities

Email Notifications

- Registration Success
 - Email Verification
 - Forgot Password
 - Password Reset
 - Assessment Assigned
 - Payment Success
 - Subscription Expiry Reminder (Optional)
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Complete Module Summary

Module	Major Functionalities
Authentication	Login, Register, JWT, Password Management
Organization	Organization & HR Management
Subscription	Plans, Payments, Razorpay
Candidate Management	Candidate CRUD & Profile
Question Bank	MCQ & Coding Question Management
Assessment	Assessment Creation & Publishing
Candidate Assignment	Assign Exams & Notifications
Online Examination	Exam Flow, Timer, Submission
Coding Editor	Monaco Editor, Run & Submit Code
AI Code Evaluation	Judge0, LLM Evaluation, AI Feedback

AI Proctoring	Browser Monitoring, Camera Monitoring, Event-Based Video Recording, Violation Logging
Results	Scores, Pass/Fail, AI Scores
Reports	Candidate, AI & Proctoring Reports
Notification	Email Notifications